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Power tool

Abstract

A power tool includes an output assembly and a battery pack. The output assembly is disposed at an end portion of a first handle, the output assembly includes at least two output units spaced apart and used for connecting working accessories, and the output assembly is rotatable about a shift axis to shift an output unit among the at least two output units to a working position or a non-working position. The battery pack is disposed at a bottom of a body, the battery pack includes a standing surface through which the power tool is capable of standing, and when the power tool stands, a projection of a center of gravity of the power tool on the standing surface is in the standing surface.

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Background/Summary

RELATED APPLICATION INFORMATION (1) This application is a continuation of International Application Number PCT/CN2021/101437, filed on Jun. 22, 2021, through which this application also claims the benefit under 35 U.S.C. § 119(a) of Chinese Patent Application No. CN 202010631586.7 filed Jul. 3, 2020, Chinese Patent Application No. CN 202011158306.1 filed Oct. 26, 2020, Chinese Patent Application No. CN 202110625432.1 filed Jun. 4, 2021, and Chinese Patent Application No. CN 202110624692.7 filed Jun. 4, 2021, which are incorporated by reference in their entireties.

TECHNICAL FIELD

(1) The present application relates to the technical field of power tools, for example, a power tool with multiple output units.

BACKGROUND

(2) Typically, a handheld power tool has only one output head for mounting a working accessory. Therefore, when the power tool is used, the working accessory needs to be replaced in conjunction with different use conditions. When the working accessory is replaced, the original working accessory needs to be removed and then a working accessory to be used is mounted. It is cumbersome to remove and mount the working accessories, which affects a user's experience and work efficiency in a working process. Therefore, some power tools each of which has double working heads appear in the related art, where positions of the double working heads are shifted so that requirements of different working conditions can be met.

(3) However, a power tool with double working heads in the related art has the following problems: firstly, since a working head and a structure for shifting the double working heads are added, the power tool with the double working heads typically has a complicated structure and a great weight, which is inconvenient for the user to operate, has redundant structures occupying a relatively large space of the whole machine, and is not conducive to the miniaturization of the whole machine; secondly, the arrangement of internal structures and the shape of the whole machine are improper, resulting in the deviation of the center of gravity of the whole machine, which causes the tool to stand unsteadily during use, leads to poor user experience, and reduces the work efficiency of the tool; and thirdly, a clutch mechanism is typically further provided so that it is convenient to break a transmission connection between a working head and a transmission mechanism when the working head is shifted, a locking switch in the related art for implementing a disconnection and a connection is typically disposed on a handle of a main body and far from the working head, and when a shift operation is performed, the user needs to press the locking switch with one hand for an unlocking purpose and rotate the working heads with the other hand to shift the working heads, which is not conducive to a shift by the user with a single hand and affects the operation experience of the user.

SUMMARY

(4) An example provides a power tool including a body, an output assembly, a power mechanism, a switch assembly, and a battery pack. The body includes a first handle. The output assembly is disposed at an end portion of the body. The power mechanism includes an electric motor and a transmission unit which drive the output assembly to move. The switch assembly is disposed on the body and controls the power mechanism. The battery pack supplies power to the power mechanism. The output assembly includes a first output unit and a second output unit, where the first output unit and the second output unit are spaced apart, the first output unit is used for connecting a first working accessory, the second output unit is used for connecting a second working accessory, and the output assembly is rotatable about a shift axis to shift the first output unit to a working position or a non-working position. The battery pack is disposed at a bottom of the body, the battery pack includes a standing surface through which the power tool is capable of standing, and when the power tool stands, an orthographic projection of a center of gravity of the power tool on the standing surface is in the standing surface.

(5) An example provides a power tool including a body, an output assembly, a power mechanism, a switch assembly, and a battery pack. The body includes a first handle. The output assembly is disposed at an end portion of the first handle. The power mechanism includes an electric motor and a transmission unit which drive the output assembly to move. The switch assembly is disposed on the body and controls the power mechanism. The battery pack supplies power to the power mechanism. The output assembly includes at least two output units spaced apart and used for connecting working accessories. The power tool further includes a shift mechanism including

mounting portions, where each of the at least two output units is supported by a respective one of the mounting portions, and the shift mechanism is rotatably disposed at the end portion of the first handle about a shift axis. The battery pack is disposed at a bottom of the body, the battery pack includes a standing surface through which the power tool is capable of standing, and when the power tool stands, an orthographic projection of a center of gravity of the power tool on the standing surface is in the standing surface.

(6) An example provides a power tool including a body, an output assembly, a power mechanism, and a switch assembly. The body includes a first handle. The output assembly is disposed at an end portion of the first handle. The power mechanism includes an electric motor and a transmission unit which drive the output assembly to move. The switch assembly is disposed on the body and controls the power mechanism. The output assembly includes at least two output units spaced apart and used for connecting working accessories. The power tool further includes a shift mechanism and a clutch mechanism. The shift mechanism includes mounting portions, where each of the at least two output units is supported by a respective one of the mounting portions, and the shift mechanism is rotatably disposed at the end portion of the first handle about a shift axis. The clutch mechanism is movably connected to the shift mechanism, where the clutch mechanism has a transmission position where an output unit among the at least two output units is in a transmission connection to the transmission unit and a disengagement position where the clutch mechanism is disengaged from the shift mechanism, where when the clutch mechanism is in the disengagement position, the shift mechanism is rotatable with respect to the body.

(7) An example provides a power tool including a body, an output assembly, a power mechanism, and a switch assembly. The body includes a first handle. The output assembly is disposed at an end portion of the first handle. The power mechanism includes an electric motor and a transmission unit which drive the output assembly to move. The switch assembly is disposed on the body and controls the power mechanism. The output assembly includes at least two output units spaced apart and used for connecting working accessories, and the output assembly is rotatable about a shift axis to shift an output unit among the at least two output units to a working position or a non-working position. The power tool further includes a clutch mechanism movably connected to a shift mechanism, where the clutch mechanism has a transmission position where the output unit is in a transmission connection to the transmission unit and a disengagement position where the clutch mechanism is disengaged from the output assembly, where when the clutch mechanism is in the disengagement position, the output assembly is rotatable with respect to the body.

(8) In some examples, the first handle is supportively connected between the output assembly and the battery pack.

(9) In some examples, the first handle is disposed obliquely with respect to the shift axis.

(10) In some examples, an included angle between an axis of the first handle and the shift axis is greater than or equal to 10° and less than or equal to 60° .

(11) In some examples, an included angle between an axis of an output unit and the standing surface is greater than or equal to 0° and less than or equal to 90° .

(12) In some examples, the output unit in the working position is higher than the output unit in the non-working position.

(13) In some examples, an axis of the first handle and the shift axis are in different planes and form a cross shape.

(14) In some examples, the electric motor is disposed in the first handle.

(15) In some examples, the switch assembly includes a trigger for controlling the electric motor to start or stop.

(16) In some examples, the trigger includes a front limit position when the trigger is not operated, where an operation span between the front limit position and a front end portion of an output unit among the at least two output units along an axial direction of the output unit is less than or equal to 130 mm.

(17) In some examples, a span of a whole machine between a rear limit position of the body and the front end portion of the output unit along the axial direction of the output unit is less than or equal to 195 mm.

(18) In some examples, the power tool further includes the shift mechanism including mounting portions, where each of the at least two output units is supported by a respective one of the mounting portions, and the shift mechanism is rotatably disposed at the end portion of the first handle.

(19) In some examples, a mounting portion is a U-shaped support including a connection base and a pair of sidewalls, where the connection base is connected between the pair of sidewalls.

(20) In some examples, the shift mechanism further includes a shift unit connected to the mounting portions, and the shift unit is an annular frame rotatable about the shift axis.

(21) In some examples, the power tool further includes the clutch mechanism movably connected to the shift mechanism, where the clutch mechanism has the transmission position where the output unit is in the transmission connection to the transmission unit and the disengagement position where the clutch mechanism is disengaged from the output unit, where when the clutch mechanism is in the disengagement position, the output unit is rotatable with respect to the body.

(22) In some examples, the shift mechanism guides the clutch mechanism to move between the transmission position and the disengagement position.

(23) In some examples, the clutch mechanism or the transmission unit is at least partially disposed between two output units spaced apart.

(24) In some examples, the clutch mechanism includes a shaft sleeve and a moving member, where the moving member is connected to the shaft sleeve and the shaft sleeve is movably connected to the transmission unit and driven by the moving member to be connected to or disengaged from the output unit.

(25) In some examples, the transmission unit includes a transmission shaft, where the transmission shaft includes a clutch transmission portion, the shaft sleeve is movably connected to the clutch transmission portion, and the transmission shaft transmits torque to the shaft sleeve via the clutch transmission portion.

(26) In some examples, the clutch transmission portion includes a noncircular shaft disposed on an end portion of the transmission shaft, and the noncircular shaft mates with and is connected to a noncircular hole on the shaft sleeve.

(27) In some examples, the output unit, the shaft sleeve, and the transmission shaft are disposed coaxially, the transmission position and the disengagement position are distributed along an axial direction of the transmission shaft, and the moving member drives the shaft sleeve to move along the axial direction of the transmission shaft.

(28) In some examples, the shift mechanism includes a locking portion and an unlocking portion, where when the clutch mechanism is in the transmission position, the clutch mechanism mates with the locking portion and restrains the shift mechanism from rotating, and when the clutch mechanism is in the disengagement position, the clutch mechanism is disengaged from the locking portion and allows the shift mechanism to rotate.

(29) In some examples, the locking portion includes multiple straight slots, the unlocking portion is an annular slot, the annular slot and the shift axis are disposed coaxially, and an end portion of each of the multiple straight slots intersects with the annular slot.

(30) In some examples, the moving member includes a first connection portion adapted to be inserted into a straight slot and an annular slot, where the annular slot forms the disengagement position and the straight slot forms the transmission position.

(31) In some examples, the shift mechanism further includes a shift unit connected to the mounting portions, and the shift unit is an annular frame rotatable about the shift axis.

(32) In some examples, the shift unit is the annular frame rotatable about the shift axis.

(33) In some examples, a locking portion includes locking slots distributed along a radial direction

of the annular frame, where one of the locking slots causes the clutch mechanism to be in the transmission position.

(34) In some examples, the shift mechanism is a shell-shaped structure and provided with an opening at a bottom of the shift mechanism.

(35) In some examples, the shift unit is disposed at the opening.

(36) In some examples, the shift mechanism is a U-shaped support including a connection base and a pair of sidewalls, where the connection base is connected between the pair of sidewalls.

(37) In some examples, the clutch mechanism or the transmission unit is at least partially disposed between the pair of sidewalls.

(38) In some examples, a torque transmission member is provided at another end of the shaft sleeve, and the output unit includes a clamping portion and a connection portion, where the clamping portion is used for mounting a working accessory, and the connection portion is in the transmission connection to the torque transmission member.

(39) In some examples, the torque transmission member includes engagement slots disposed on the shaft sleeve, and the connection portion includes a transmission claw disposed on the output unit, where the engagement slots are adapted for insertion of the transmission claw.

(40) In some examples, the body further includes a support housing disposed at a top of the first handle, where the support housing includes an output port and a rotation guide unit, and a mounting portion is slidably connected to the rotation guide unit.

(41) In some examples, the power tool further includes a protective housing connected to the body, where a protective space is formed between the protective housing and the body and used for accommodating a working accessory in the non-working position.

(42) In some examples, the transmission unit further includes a first bevel gear and a second bevel gear which mesh with each other, where the first bevel gear is coaxially connected to the transmission shaft.

(43) In some examples, the body further includes a second handle, where the second handle is disposed at a rear end of the first handle.

(44) In some examples, the battery pack is disposed at a bottom of the first handle and a bottom of the second handle, the battery pack includes the standing surface through which the power tool is capable of standing, and when the power tool stands, the orthographic projection of the center of gravity of the power tool on the standing surface is in the standing surface.

(45) In some examples, a cross-section of the first handle and a cross-section of the second handle at any same height have unequal dimensions, where a height direction refers to a direction perpendicular to the standing surface.

(46) In some examples, the switch assembly is disposed on the second handle.

(47) In some examples, the power tool further includes a storage mechanism for storing unused accessories, where the storage mechanism is a storage bin or a storage clip, the storage bin is disposed in the body and has a lid which is openable and closeable, and the storage clip has slots for clamping the accessories.

(48) The power tool in the examples of the present application has a proper center of gravity and is stable during use, structures inside a housing are compact, and the tool is more miniature. In the present application, operation steps of a user are simplified so that the user can perform a disconnection, a connection, and a shift with a single hand, improving user experience.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a structure view of a power tool in the present application;

(2) FIG. 2 is a schematic view showing the assembly of the power tool shown in FIG. 1;

- (3) FIG. 3 is a schematic view showing the assembly of the power tool shown in FIG. 1 from another angle;
- (4) FIG. 4 is a structure view of the power tool shown in FIG. 2 with part of a housing removed;
- (5) FIG. 5 is a schematic view showing the assembly of output units, a shift mechanism, a clutch mechanism, and a transmission unit in the present application;
- (6) FIG. 6 is a front view of a shift mechanism in the present application;
- (7) FIG. 7 is a bottom view of the shift mechanism shown in FIG. 6;
- (8) FIG. 8 is a front view of a moving member in the present application;
- (9) FIG. 9 is a left view of the moving member shown in FIG. 8;
- (10) FIG. 10 is a top view of the moving member shown in FIG. 8;
- (11) FIG. 11 is a structure view of another example of a shift mechanism in the present application;
- (12) FIG. 12 is a schematic view of output units, the shift mechanism, and a clutch mechanism in the example in FIG. 11, which are assembled;
- (13) FIG. 13 is a structure view showing the mounting of the shift mechanism in the example in FIG. 11;
- (14) FIG. 14 is a structure view showing that the shift mechanism causes a clutch mechanism to be in a transmission position in the example in FIG. 11;
- (15) FIG. 15 is a structure view showing that the shift mechanism causes a clutch mechanism to be in a disengagement position in the example in FIG. 11;
- (16) FIG. 16 is a schematic view showing the assembly of the shift mechanism and output units in the example in FIG. 11;
- (17) FIG. 17 is a schematic view showing the assembly of a clutch mechanism in the example in FIG. 11;
- (18) FIG. 18 is a structure view of a shift unit of the shift mechanism in the example in FIG. 11,
- (19) FIG. 19 is a structure view of an output unit in the present application;
- (20) FIG. 20 is a front view of the output unit shown in FIG. 19;
- (21) FIG. 21 is a structure view of a shaft sleeve in the present application;
- (22) FIG. 22 is a left view of the shaft sleeve shown in FIG. 21; and
- (23) FIG. 23 is a structure view of a storage mechanism in the present application.

DETAILED DESCRIPTION

- (24) As shown in FIG. 1, the present application provides a power tool **100**, for example, a handheld power tool, where the power tool **100** in an example of the present application is a handheld dual-output electric drill.
- (25) In addition, although this example relates to the handheld power tool, it is to be understood that the present application is not limited to the disclosed examples but is applicable to other types of power tools, including, but not limited to, other tools which need to clamp working accessories, such as an electric drill.
- (26) As shown in FIG. 1, the power tool **100** includes a body **110**, an output assembly **200**, a power supply device, and a switch assembly **800**. The power supply device in the example of the present application is a battery pack **700**. In other examples, the power tool **100** may be an alternating current tool, and in this case, the power supply device includes a cable and a plug which are connected to mains electricity.
- (27) As shown in FIG. 1, the body **110** in the example of the present application includes a first handle **111** and a second handle **112**, where the first handle **111** and the second handle **112** are disposed along a front and rear direction of the body **110**, and the second handle **112** is disposed at a rear end of the first handle **111**. The front and rear direction mentioned in this example refers to that when the power tool is gripped, a direction which a user faces is the front, and a direction opposite to the direction which the user faces is the rear.
- (28) A cross-section of the first handle **111** and a cross-section of the second handle **112** at any same height have unequal dimensions, where a height direction refers to a direction perpendicular

to a standing surface.

(29) In this example, the battery pack **700** is disposed at a bottom end of the first handle **111** and a bottom end of the second handle **112**, and optionally, the bottom end of the first handle **111** and the bottom end of the second handle **112** are connected through the battery pack. A connection base **113** adapted for mounting the battery pack **700** is further provided at the bottom end of the first handle **111** and the bottom end of the second handle **112**, and conductive terminals electrically connected to the battery pack **700** are disposed on the connection base **113**. The battery pack **700** includes a standing surface **710**, where the tool **100** may stand on a surface of a worktable through the standing surface **710** of the battery pack **700**, and in the example of the present application, an orthographic projection of a center of gravity of the power tool **100** on the standing surface **710** is in the standing surface **710**. With the preceding arrangement, the case can be avoided where the tool stands unsteadily due to an unsteady center of gravity of the tool, thereby avoiding a safety hazard caused by the fall of the tool. When the tool stands on the worktable, part of a bottom surface of the battery pack **700** is in contact with an operation table and constitutes the standing surface **710**. In an example, if the bottom surface of the battery pack **700** is a plane, the bottom surface of the battery pack **700** constitutes the standing surface **710**.

(30) The output assembly **200** in the example of the present application is disposed at a front end of the body **110**, where the output assembly **200** includes at least two output units **210** spaced apart and used for connecting working accessories **900**. Referring to FIG. 1, an output unit **210** has a working position **120** and a non-working position **130**, where the working position **120** is on a front side of the output assembly **200**, and any other position except the front side may constitute the non-working position. The output assembly **200** has a shift axis **201** about which the output assembly **200** rotates, where the output assembly **200** rotates about the shift axis **201** and can rotate different output units **210** on the output assembly **200** to the working position **120** or the non-working position **130**.

(31) As shown in FIGS. 2 and 3, the output unit **210** in this example is in the shape of a cylinder whose axis coincides with an output axis **202**. In this example, two output units **210** are disposed opposite to each other. Therefore, in the example of the present application, the front side of the output assembly **200** is the working position **120**, and a rear side corresponding to the front side is the non-working position **130**. When one of the working accessories **900** needs to be used, the output unit **210** corresponding to the one of the working accessories **900** is rotated to the front side. An end portion of the output unit **210** is provided with a clamping portion **211** for clamping a working accessory **900**, where the working accessory **900** in this example includes a drill bit, a screwdriver bit, and the like. One of the two output units **210** is configured to be used for clamping the drill bit, and the other output unit **210** is configured to be used for clamping the screwdriver bit. As an alternative example, the two output units **210** may be used for clamping two drill bits or two screwdriver bits, which is not limited.

(32) Referring to FIG. 4, a power mechanism **300** and a control mechanism **400** are disposed in the body **110**, where the power mechanism **300** is used for driving the output unit **210** to move, the control mechanism **400** is used for controlling the running of the power mechanism **300**, the power mechanism **300** includes an electric motor **310** and a transmission unit **320**, and an output end of the electric motor **310** is in a transmission connection to the output unit **210** through the transmission unit **320**.

(33) As shown in FIG. 1, the output unit **210** has the output axis **202** about which the output unit **210** rotates, where the two output units in the example of the present application are disposed coaxially along a direction of the output axis **202**, and the output axis **202** is disposed obliquely with respect to the standing surface **710**. For example, as shown in FIG. 1, the output unit **210** in the working position **120** is higher than the output unit **210** in the non-working position **130**. In this example, referring to FIG. 1, the output axis **202** is approximately perpendicular to the shift axis **201**, and the output unit **210** extends in a direction approximately perpendicular to the shift axis

201. In other words, the output unit **210** rotates in a rotation plane approximately perpendicular to the shift axis **201**, and the rotation plane is disposed obliquely with respect to the standing surface **710**.

(34) In the example of the present application, an axis of the first handle **111** and the shift axis **201** are configured to intersect with each other. For example, the axis of the first handle **111** and the shift axis **201** may be in different planes and form a cross shape or may intersect with each other in the same plane. When the axis of the first handle **111** and the shift axis **201** are in the different planes and form the cross shape, an included angle α between the axis of the first handle **111** and the shift axis **201** is greater than or equal to 0° and less than or equal to 60° . When the axis of the first handle **111** and the shift axis **201** intersect with each other in the same plane, the included angle α between the axis of the first handle and the shift axis is greater than or equal to 10° and less than or equal to 60° .

(35) Referring to FIG. 4, the electric motor **310** is disposed in the first handle **111**, where the first handle **111** is supportively connected between the output assembly **200** and the battery pack **700** and disposed obliquely with respect to the shift axis **201**. Referring to FIG. 2, in the example of the present application, the included angle α between the axis of the first handle **111** and the shift axis **201** is greater than or equal to 10° and less than or equal to 60° , and an included angle β between the output axis **202** of the output unit **210** and the standing surface **710** is greater than or equal to 0° and less than or equal to 90° . The output axis **202** is disposed obliquely with respect to the standing surface **710** and the included angle between the first handle **111** and the shift axis **201** is set in the preceding range so that the first handle is longer than the second handle, thereby placing most mechanisms in the first handle. On the one hand, on the basis that transmission performance is ensured, a dimension of the whole machine in an axial direction is reduced as much as possible so that the whole machine has a more compact structure and a smaller dimension. On the other hand, the first handle **111** is disposed obliquely and the electric motor **310** is disposed in the first handle **111** so that the electric motor **310** is closer to the output unit **210**, and the center of gravity of the whole machine is adjusted forward through the weight of the electric motor **310** and finally the orthographic projection of the center of gravity on the standing surface **710** is in the standing surface **710** so that the whole machine is more harmonious and steadier. In other examples, as long as the projection of the center of gravity of the tool **100** on the standing surface **710** is in the standing surface **710**, the first handle **111** may be perpendicular to the standing surface **710**, and in this case, the shift axis **201** is also oblique with respect to the standing surface **710**.

(36) The control mechanism **400** mainly includes a control circuit board and is electrically connected to the switch assembly **800** and the electric motor **310** separately. The switch assembly **800** controls, through the control mechanism **400**, the electric motor **310** to be on or off to switch a working state of the whole machine. The switch assembly **800** is disposed on the body **110**, for example, the switch assembly **800** is disposed on the second handle **112**, and the control mechanism **400** is disposed in the connection base **113**. The control mechanism **400** is disposed in the connection base **113** so that the control mechanism is in the middle of the switch assembly **800**, the electric motor **310**, and the battery pack **700**, which makes circuit arrangement easier, saves a space occupied by lines, and further reduces the dimension of the whole machine.

(37) Referring to FIG. 4, the switch assembly **800** includes a trigger **810** and a locking button **820**, where the trigger **810** is electrically connected to the control mechanism **400** and used for controlling the electric motor **310** to start or stop. The trigger **810** has an original position where the trigger **810** is not operated by the user and an activation position where the trigger **810** is pressed to activate the electric motor **310**, where the locking button **820** may be used for locking the trigger **810** in the activation position. As shown in FIG. 4, an operation span $L1$ of the trigger **810** is less than or equal to 130 mm, where the operation span refers to a distance between a forefront of the trigger **810** in the original position **811** and a front end portion of the output unit **210** along an axial direction of the output unit. In an example, a span $L2$ of the whole machine between a rear limit

position of the body and the front end portion of the output unit **210** is less than or equal to 195 mm, where the span **L2** of the whole machine refers to a distance between a rear end portion of the body and the front end portion of the output unit **210** along the axial direction of the output unit. In this example, the span **L2** of the whole machine is approximately 175 mm, and the height of the whole machine is about 167 mm, where the height refers to a longitudinal distance from a top end of the body to the standing surface **710**.

(38) If the operation span **L1** is excessively long, the working accessory **900** is liable to have an unsteady state such as a swing in a working process, which increases the difficulty of the user in manipulation and is not conducive to operation of the user with a single hand. Therefore, the operation span is set in the preceding range, which is conducive to reducing the difficulty of the user in manipulation, facilitates the operation of the user with a single hand, and improves user experience.

(39) The power tool in the example of the present application further includes a shift mechanism **500** and a clutch mechanism **600**. As shown in FIGS. **1** to **4**, the shift mechanism **500** is rotatably disposed at a top of the first handle **111** and rotates about the shift axis **201**, where the clutch mechanism **600** is movably connected to the shift mechanism **500**. In other examples of the present application, the shift mechanism may not be provided, and the output assembly actually has a shift function. That is, the output assembly and the shift mechanism are integrated so that the output assembly includes a structure which can implement locking and unlocking actions with the clutch mechanism.

(40) The shift mechanism **500** includes mounting portions **510** and a locking element, where a mounting portion **510** is used for mounting and supporting the output unit **210**, and the locking element can implement the movement of the clutch mechanism **600** between a transmission position and a disengagement position.

(41) As shown in FIGS. **5** to **7**, in an example, the locking element includes a locking portion **520** and an unlocking portion **530**, where the locking portion **520** restrains the shift mechanism **500** from rotating with respect to the clutch mechanism **600**, and the unlocking portion **530** allows the shift mechanism **500** to rotate.

(42) Referring to FIG. **7**, the clutch mechanism **600** has a transmission position **601** where the output unit **210** is in the transmission connection to the transmission unit **320** and a disengagement position **602** where the transmission unit **320** is disengaged from the output unit **210**, and the shift mechanism **500** guides the clutch mechanism **600** to move between the transmission position **601** and the disengagement position **602**, where the transmission position **601** and the disengagement position **602** are distributed along an axial direction of a transmission shaft **321**.

(43) As shown in FIGS. **5** and **6**, the shift mechanism in the example of the present application is a U-shaped support including a connection base **540** and a pair of sidewalls **550**, where the connection base **540** is disposed between the pair of sidewalls **550** such that U-shaped openings **545** opposite to each other are formed between the pair of sidewalls. The connection base **540** is pivotally connected to a support housing **114** at a top of the body **110**, where the support housing **114** is disposed at the top of the first handle **11**. Referring to FIGS. **2** and **3**, the support housing **114** includes an output port **1141** via which the clutch mechanism **600** is connected to the transmission shaft **321**, and the shift mechanism **500** is pivotally connected to the support housing **114**.

(44) As shown in FIGS. **5** and **6**, the mounting portion **510** is a support boss disposed on a sidewall **550**, where a circular mounting hole is disposed on the support boss and penetrates through the support boss to fit the output unit **210**, and the output unit **210** is supported in the mounting hole of the support boss.

(45) As shown in FIG. **5**, the clutch mechanism **600** includes a shaft sleeve **610** and a moving member **620**, where the moving member **620** is connected to the shaft sleeve **610**, and the shaft sleeve **610** is movably connected to the transmission shaft **321** and may be connected to or disengaged from the output unit **210**. The output unit **210**, the shaft sleeve **610**, and the

transmission shaft **321** are disposed coaxially, and the moving member **620** drives the shaft sleeve **610** to move along the axial direction of the transmission shaft **321**.

(46) Referring to FIGS. **8** to **10**, the moving member **620** in the example of the present application includes a first connection portion **621** and a second connection portion **622**, where the first connection portion **621** is slidably connected to the locking portion **520** and the unlocking portion **530**, and the locking portion **520** guides the moving member **620** to move between the transmission position **601** and the disengagement position **602**; and the second connection portion **622** is connected to the shaft sleeve **610** and the shaft sleeve **610** is driven to move synchronously through the second connection portion **622**.

(47) As shown in FIG. **3**, the moving member **620** is disposed across the shaft sleeve **610** along a radial direction. Referring to FIGS. **8** to **10**, the moving member **620** includes a moving member body **623**, the first connection portion **621** is a protrusion disposed at a top of the moving member body **623**, and the second connection portion **622** is a claw disposed below the moving member body **623**. As shown in FIG. **21**, a groove **612** is disposed on the shaft sleeve **610** corresponding to the claw, for example, the groove **612** is an annular groove cut along an outer circumference of the shaft sleeve **610**, where the claw is clamped in the groove **612**.

(48) Referring to FIGS. **8** to **10**, the moving member **620** in the example of the present application is further provided with operation portions **624** for the user to operate, where two operation portions are provided on two radial sides of the support housing **114** separately and exposed via the openings **545** of the shift mechanism **500** to be operated by the user, and the user operates the operation portions **624** to shift the clutch mechanism **600**.

(49) As shown in FIG. **2**, the clutch mechanism **600** in the example of the present application further includes a biasing member **630** disposed at an end portion of the moving member **620**, where after the user releases the operation portions **624**, the moving member **620** can return to the transmission position **601** under the action of the biasing member **630**.

(50) As shown in FIG. **7**, in the example of the present application, the locking portion **520** and the unlocking portion **530** are each a slot disposed on an inner side of the connection base **540**, where the first connection portion **621** is adapted to be inserted into the slot and move along the slot. The locking portion **520** includes multiple straight slots disposed along an axial direction of the mounting portion **510**, and the unlocking portion **530** is an annular slot disposed in the center of the connection base **540**, where the straight slots extend along the axial direction of the mounting portion **510** separately, and the annular slot and the shift axis **201** are disposed coaxially. An end portion of each of the multiple straight slots intersects with the annular slot, where the disengagement position **602** is from an intersection between a straight slot at one end and the annular slot, through the annular slot, and to an intersection between a straight slot at the other end and the annular slot, and an end of the straight slot facing away from the annular slot forms the transmission position **601**.

(51) When the first connection portion **621** is at the end of the straight slot facing away from the annular slot, the clutch mechanism **600** is in the transmission position **601**. In this case, the shaft sleeve **610** is driven by the moving member **620** to extend out via the output port **1141** of the support housing **114** and be connected to the output unit **210**. Since the protrusion is in the straight slot, the shift mechanism **500** is restrained by the straight slot and the clutch mechanism from rotating with respect to the clutch mechanism **600**. When the protrusion is in the position where the straight slot intersects with the annular slot, in other words, the protrusion is in the annular slot, the clutch mechanism **600** is in the disengagement position **602**, and the shaft sleeve **610** is driven by the moving member **620** to be retracted into the support housing **114** and disengaged from the output unit **210**. Since the protrusion is in the annular slot, the shift mechanism **500** may rotate with respect to the clutch mechanism **600**.

(52) As shown in FIG. **3**, a rotation guide unit **1142** is further disposed on the support housing **114**, and the shift mechanism **500** is slidably connected to the rotation guide unit **1142**. For example, as

shown in the figure, the rotation guide unit **1142** in the example of the present application is a slide slot disposed on the support housing **114**, and bottoms of the pair of sidewalls **550** of the shift mechanism **500** are slidably connected to the slide slot.

(53) As shown in FIG. **4**, the transmission unit **320** includes a gearbox including a bevel gearset **322** and the transmission shaft **321**. Since the first handle **111** in this example is disposed obliquely with respect to the standing surface **710**, the gearbox optionally includes the bevel gearset. In other examples, the gearbox may include any other type of transmission structure, which is not limited. The bevel gearset **322** includes a first bevel gear and a second bevel gear, where the first bevel gear and the transmission shaft **321** are coaxially connected to each other and disposed together between the two output units spaced apart.

(54) As shown in FIG. **5**, the transmission shaft **321** includes a clutch transmission portion **3211**, where the shaft sleeve **610** is slidably connected to the clutch transmission portion **3211**, and the transmission shaft **321** transmits torque to the shaft sleeve **610** via the clutch transmission portion **3211** such that the shaft sleeve **610** rotates synchronously with the transmission shaft **321**.

(55) Referring to FIG. **5**, the clutch transmission portion is a noncircular shaft disposed on an end portion of the transmission shaft **321**, that is, the clutch transmission portion has a noncircular cross-section, and a noncircular hole mating with the noncircular shaft is disposed on the shaft sleeve **610** corresponding to the clutch transmission portion. The transmission shaft **321** in this example includes a flat shaft **3211** in the transmission connection to the shaft sleeve **610**, and a flat hole **613** mating with the noncircular shaft is disposed on the shaft sleeve **610** corresponding to the transmission shaft **321**. The flat shaft **3211** mates with the flat hole **613** such that the shaft sleeve **610** is allowed to axially slide with respect to the transmission shaft **321** or rotate with the transmission shaft **321**. As an alternative example, the clutch transmission portion **3211** may be a plane disposed on an outer circumference of the transmission shaft **321**, that is, the plane may be cut on the outer circumference of the transmission shaft **321** along the axial direction of the transmission shaft **321** as long as the transmission shaft **321** forms a noncircular cross-section, and a hole having the same shape as the noncircular cross-section is disposed on the shaft sleeve.

(56) A torque transmission member is provided at the other end of the shaft sleeve **610** in the example of the present application, and a connection portion is provided at the other end of the output unit **210** without the clamping portion, where the connection portion in the transmission connection to the torque transmission member. For example, as shown in FIGS. **20** and **21**, the connection portion is a transmission claw **212** disposed on the end portion of the output unit **210**, and the torque transmission member is engagement slots **611** disposed on an inner circumferential surface of the shaft sleeve **610**, where the engagement slots **611** are disposed in correspondence with the transmission claw **212** and adapted for the insertion of the transmission claw **212**, thereby implementing the rotation of the output unit **210** with the shaft sleeve **610**.

(57) As shown in FIGS. **1** to **3**, the power tool **100** in the example of the present application further includes a protective housing **115** connected to the body **110**, where a protective space **1151** is formed between the protective housing **115** and the body **110** and used for accommodating and protecting the working accessory **900** in the non-working position. In an example, the protective housing **115** is disposed only at a top of the output assembly, where one end of the protective housing **115** is pivotally connected to a shift shaft of the shift mechanism **500**, and the other end of the protective housing **115** is connected to the body. The protective housing **115** has a stepped shape to fit the shape of the shift mechanism and the shape of the working accessory in the non-working position so that the overall dimension of the housing is more compact.

(58) As shown in FIGS. **3** and **4**, the rotation guide unit **1142** is further disposed on the support housing **114**, and the shift mechanism **500** is slidably connected to the rotation guide unit **1142**. Optionally, the rotation guide unit **1142** in the example of the present application is the slide slot disposed on the support housing **114**, and circular plate stiffeners slidably connected to the slide slot are formed at a bottom of the connection base **540** of the shift mechanism **500**.

(59) As shown in FIG. 23, the power tool in the present application further includes a switching key **830** for switching an output rotational speed of the electric motor. The switching key **830** is disposed on the first handle **111**, for example, the switching key **830** is disposed on a side of the first handle **111** facing the second handle **112**, where the side of the first handle **111** facing the second handle **112** is equivalent to a side of the first handle **111** facing the user, thereby facilitating the operation of the user. The switching key **830** is integrated with indicator lights indicating a current state of the rotational speed of the electric motor. In this example, the electric motor has at least two switchable rotational speeds, and at least two corresponding indicator lights are provided, where the indicator lights have different lengths for distinguishing different rotational speeds of the electric motor. In other examples, colors of the indicator lights may be used for distinguishing the different rotational speeds of the electric motor.

(60) As shown in FIG. 23, the power tool in the example of the present application further includes a storage mechanism for storing an accessory **910** to be used, a replaced accessory **910**, or a spare accessory **910**, where the storage mechanism is a storage bin **920** or a storage clip **930**.

(61) The storage bin **920** is disposed in a handle and has an openable lid and a storage space **921**. In this example, the storage bin is disposed on a rear side of the second handle **112**, the lid is opened toward the user, and the accessory **910** is placed in an engagement slot on an inner side of the lid or disposed in the storage space **921**. In other examples, the storage bin **920** may be the preceding structure disposed in another position, a drawer-shaped structure, or the like, which is not limited herein.

(62) The storage clip **930** is disposed on a surface of the housing, where the storage clip **930** has multiple slots where the accessories **910** are placed, and the accessories **910** are engaged in the slots. In an example, the storage clip **930** is made of soft rubber and disposed below the second handle **112** and behind the connection base **113**.

(63) When the power tool is used normally, the moving member **620** is in the transmission position **601**, and the shaft sleeve **610** is in the transmission connection to the output unit **210** at the front end. In this case, the torque of the transmission shaft **321** of the gearbox may be normally transmitted to the output unit **210**.

(64) When the working accessory **900** needs to be shifted, the user manually pushes the operation portions **624** of the clutch mechanism **600** to cause the moving member **620** to move toward the rear end along the straight slot, and the shaft sleeve **610** is disengaged from the output unit **210** in the working position. When the moving member **620** moves to a top end of the straight slot and cannot continue to move, the moving member **620** is in the annular slot (that is, the moving member **620** is in the disengagement position), and at this time, the user may manually rotate the shift mechanism **500** to rotate the output unit in the working position to the non-working position and rotate the output unit in the non-working position to the working position.

(65) When the user releases the operation portions **624**, the moving member **620** automatically returns to the transmission position under the action of the biasing member, the shaft sleeve **610** is in the transmission connection to the output unit in the working position, and the output unit may work normally.

(66) In the present application, the preceding shift mechanism and the preceding clutch mechanism which are linked to each other are provided such that the clutch mechanism and the shift mechanism are configured to collaborate. The clutch mechanism is in the transmission connection to the body and an output mechanism so that the output is transmitted or terminated. In addition, the clutch mechanism and the shift mechanism are linked to each other through the first connection portion, the locking portion, and the unlocking portion so that a shift and a rotation are allowed to be performed while a connection and a disconnection are performed. Thus, operation steps of the user are simplified so that the user can perform the disconnection, the connection, and the shift with a single hand, improving the user experience.

(67) In another example of the present application, the locking element includes a shift unit **570**,

where the shift unit **570** includes a locking slot **571** for restraining the shift mechanism **500** from rotating with respect to the body.

(68) As shown in FIGS. **11** to **18**, the shift mechanism **500** in this example is a shell structure. Optionally, the shift mechanism **500** is a T-shaped sleeve. A pair of mounting portions **510** are disposed at two ends of the T-shaped sleeve **560** along a transverse direction, where a cavity in the sleeve **560** is used for accommodating the transmission shaft, part of a transmission gear, and the shift unit **570**, and a cavity in the mounting portion **510** is used for accommodating part of the output unit **210**. The shift unit **570** may be disposed at a bottom end of the sleeve **560** and fixedly or detachably connected to the bottom end of the sleeve.

(69) In the example of the present application, a dimension of the sleeve **560** along a direction of the output axis **202** is less than or equal to 100 mm. It is to be understood that as shown in FIG. **16**, a dimension **L3** of the sleeve **560** in the front and rear direction is less than or equal to 100 mm. A dimension of the sleeve **560** along a direction perpendicular to both the output axis **202** and the shift axis **201** is less than or equal to 45 mm. It is to be understood that as shown in FIG. **16**, a maximum dimension **L4** of the sleeve **560** along a left and right direction is less than or equal to 45 mm.

(70) Referring to FIGS. **12** and **13**, the support housing **114** is disposed at the top of the first handle **111**, the support housing **114** includes the output port **1141** via which the clutch mechanism **600** is connected to the output unit **210**, and the shift mechanism **500** is pivotally connected to the support housing **114**. In an example, the sleeve **560** is sleeved outside the support housing **114** and rotatable around the support housing **114**. Therefore, the sleeve **560** and the support housing **114** are approximately disposed coaxially, and an axis of the sleeve **560** approximately coincides with the shift axis **201**.

(71) The sleeve **560** includes support bosses disposed on the mounting portions **510**, where a circular mounting hole is disposed on a support boss and penetrates through the support boss to fit the output unit **210**, and the output unit **210** is supported in the mounting hole of the support boss.

(72) As shown in FIG. **18**, the shift unit **570** is an annular frame rotatable about the shift axis **201**, where locking slots **571** are distributed in the annular frame along a radial direction of the annular frame. For example, the locking slots **571** include several slots disposed on the annular frame and distributed along the radial direction of the annular frame, and openings are disposed on radial sides of the locking slots **571** to allow the moving member **620** to be disengaged from the annular frame. The locking slot **571** in this example is a straight slot configured to radially penetrate through the annular frame, that is, openings are disposed at an inner end of the locking slot **571** and an outer end of the locking slot **571**, where the moving member **620** may move inward in the locking slot **571** along a radial direction of the locking slot **571** to an inner side of the annular frame, and thus the moving member **620** is disengaged from the annular frame to release the limit to the rotation of the annular frame. The locking slot may have an opening at the inner end. In other examples, the shift unit **570** may be in an oval shape or another shape, or the shift unit **570** may be a combined structure in which at least two arc-shaped strips are nested in a circular or arc-shaped slide slot. In this case, the locking slot may be a gap between every two arc-shaped strips.

(73) The annular frame is also provided with mounting slots **572** connected to the shift mechanism **500**. Referring to FIG. **12**, bosses **561** are disposed on an inner wall of a bottom opening of the sleeve **560**, where the bosses **561** are adapted to be engaged with the mounting slots **572**, thereby mounting and fixing the shift unit **570**.

(74) As shown in FIG. **17**, in this example, the clutch mechanism **600** includes the shaft sleeve **610** and the moving member **620**, where the moving member **620** is connected to the shaft sleeve **610**, and the shaft sleeve **610** is movably connected to the transmission shaft **321** and may be connected to or disengaged from the output unit **210**. The output unit **210**, the shaft sleeve **610**, and the transmission shaft **321** are disposed coaxially, and the moving member **620** drives the shaft sleeve **610** to move along the axial direction of the transmission shaft **321**.

(75) As shown in FIGS. 14 and 15, the clutch mechanism in the example of the present application further includes an operation member **640** which drives the moving member **620** to move, where the moving member **620** in the transmission position mates with the locking slot **571** and restrains the shift unit **570** from rotating, and the moving member **620** in the disengagement position is disengaged from the locking slot **571** and allows the shift unit **570** to rotate. For example, the operation member **640** is a button disposed at the front end of the body and below the shift mechanism **500**.

(76) The clutch mechanism **600** in this example further includes the biasing member **630** disposed at an end portion of the operation member **640**, where the biasing member **630** applies a biasing force to the operation member **640** to cause the operation member **640** to drive the clutch mechanism to be in the transmission position, and after the user releases the operation member **640**, the moving member **620** and the operation member **640** can return to the transmission position under the action of the biasing member **630**.

(77) For example, as shown in FIG. 17, the moving member **620** is disposed across the shaft sleeve **610** along the radial direction, and the moving member **620** is a shift fork including a shift lever **625** and a claw **626**, where the shift lever **625** and the claw **626** are integrally formed and the claw **626** is formed at a top end of the shift lever **625**. As shown in FIG. 13, the groove **612** is disposed on the shaft sleeve **610** corresponding to the claw **626**, for example, the groove **612** is the annular groove cut along the outer circumference of the shaft sleeve **610**, where the claw **626** is clamped in the groove **612**, and the shift lever **625** passes through the locking slot to be connected to the operation member **640**.

(78) When the power tool is used normally, the moving member **620** is in the transmission position, and the shaft sleeve **610** is in the transmission connection to the output unit **210** at the front end. In this case, the torque of the transmission shaft **321** of the gearbox may be normally transmitted to the output unit **210**.

(79) When the working accessory **900** needs to be shifted, the user manually presses the operation member **640** to drive the moving member **620** to move toward a radial inner side of the locking slot **571** along the locking slot **571**, the shaft sleeve **610** is disengaged from the output unit **210** in the working position, and the output unit **210** stops outputting power. After the moving member **620** moves until the moving member **620** is disengaged from the locking slot **571**, the moving member **620** is in the disengagement position, and the user may apply a force to the shift mechanism **500**, the output unit, or the working accessory manually or by means of an external object such as a wall or the worktable so that the shift mechanism **500** is rotated to rotate the output unit in the working position to the non-working position and rotate the output unit in the non-working position to the working position.

(80) When the user releases the operation member **640**, the operation member **640** automatically returns to an initial position under the action of the biasing member, the moving member **620** returns to the transmission position, the shaft sleeve **610** is in the transmission connection to the output unit **210** in the working position, and the output unit **210** may output power to the outside and work normally.

Claims

1. A power tool, comprising: a body comprising a first handle having a longitudinal axis; an output assembly disposed at an end portion of the first handle comprising at least two output units spaced apart and used for connecting working accessories; a power mechanism comprising an electric motor and a transmission unit which drive the output assembly to move about an output axis; a battery pack removably mountable to the body and electrically coupleable to the power mechanism, the battery pack including a standing surface, where the power tool may stand vertically on a surface of a worktable through the standing surface of the battery pack, the output axis is disposed

obliquely with respect to the standing surface; a switch assembly disposed on the body and controlling the power mechanism; a shift mechanism comprising mounting portions wherein each of the at least two output units is supported by a respective one of the mounting portions and the shift mechanism is rotatably disposed at the end portion of the first handle and has a shift axis about which the shift mechanism is rotatable; and a clutch mechanism movably connected to the shift mechanism; wherein the clutch mechanism has a transmission position in which an output unit among the at least two output units is in a transmission connection to the transmission unit and a disengagement position in which the clutch mechanism is disengaged from the shift mechanism, and wherein, when the clutch mechanism is in the disengagement position, the shift mechanism is rotatable with respect to the body, wherein an angle α between the longitudinal axis of the first handle and the shift axis is greater than or equal to 10° and less than or equal to 60° , the electric motor and the transmission unit are disposed in the first handle, and wherein the clutch mechanism comprises a shaft sleeve and a moving member, the moving member is connected to the shaft sleeve, and the shaft sleeve is movably connected to the transmission unit and driven by the moving member to be connected to or disengaged from the output unit.

2. The power tool according to claim 1, wherein the clutch mechanism or the transmission unit is at least partially disposed between two output units spaced apart.

3. The power tool according to claim 1, wherein the transmission unit comprises a transmission shaft, the transmission shaft comprises a clutch transmission portion, the shaft sleeve is movably connected to the clutch transmission portion, the transmission shaft transmits torque to the shaft sleeve via the clutch transmission portion, and the clutch transmission portion comprises a noncircular shaft disposed on an end portion of the transmission shaft and the noncircular shaft mates with and is connected to a noncircular hole on the shaft sleeve.

4. The power tool according to claim 3, wherein the output unit, the shaft sleeve, and the transmission shaft are disposed coaxially, the transmission position and the disengagement position are distributed along an axial direction of the transmission shaft, and the moving member drives the shaft sleeve to move along the axial direction of the transmission shaft.

5. The power tool according to claim 1, wherein the shift mechanism comprises a locking portion and an unlocking portion, when the clutch mechanism is in the transmission position, the clutch mechanism mates with the locking portion and restrains the shift mechanism from rotating, and, when the clutch mechanism is in the disengagement position, the clutch mechanism is disengaged from the locking portion and allows the shift mechanism to rotate.

6. The power tool according to claim 5, wherein the locking portion comprises a plurality of straight slots, the unlocking portion is an annular slot, the annular slot and the shift axis are disposed coaxially, and an end portion of each of the plurality of straight slots intersects with the annular slot.

7. The power tool according to claim 1, wherein the moving member comprises a first connection portion adapted to be inserted into a straight slot and an annular slot, the annular slot forms the disengagement position, and the straight slot forms the transmission position.

8. The power tool according to claim 1, wherein the shift mechanism further comprises a shift unit connected to the mounting portions.

9. The power tool according to claim 8, wherein the shift unit is an annular frame rotatable about the shift axis.

10. The power tool according to claim 9, wherein a plurality of locking slots are distributed along a radial direction of the annular frame and one of the plurality of locking slots causes the clutch mechanism to be in the transmission position.

11. The power tool according to claim 1, wherein the shift mechanism is a shell-shaped structure and provided with an opening at a bottom of the shift mechanism.

12. The power tool according to claim 11, wherein the shift unit is disposed at the opening.

13. The power tool according to claim 1, wherein the shift mechanism is a U-shaped support

comprising a connection base and a pair of sidewalls and the connection base is connected between the pair of sidewalls.

14. The power tool according to claim 13, wherein the clutch mechanism or the transmission unit is at least partially disposed between the pair of sidewalls.

15. The power tool according to claim 1, wherein a torque transmission member is provided at another end of the shaft sleeve, the output unit comprises a clamping portion and a connection portion, the clamping portion is used for mounting a working accessory, and the connection portion is in the transmission connection to the torque transmission member.

16. The power tool according to claim 15, wherein the torque transmission member comprises engagement slots disposed on the shaft sleeve, the connection portion comprises a transmission claw disposed on the output unit, and the engagement slots are adapted for insertion of the transmission claw.

17. A power tool, comprising: a body comprising a first handle; an output assembly disposed at an end portion of the first handle comprising at least two output units spaced apart and used for connecting working accessories and rotatable about a shift axis to shift an output unit among the at least two output units to a working position or a non-working position; a power mechanism comprising an electric motor and a transmission unit which drive the output assembly to move about an output axis; a battery pack removably mountable to the body and electrically coupleable to the power mechanism, the battery pack including a standing surface, where the power tool may stand vertically on a surface of a worktable through the standing surface of the battery pack, the output axis is disposed obliquely with respect to the standing surface; a switch assembly disposed on the body and controlling the power mechanism; and a clutch mechanism movably connected to a shift mechanism; wherein the clutch mechanism has a transmission position in which the output unit is in a transmission connection to the transmission unit and a disengagement position in which the clutch mechanism is disengaged from the output assembly and, when the clutch mechanism is in the disengagement position, the output assembly is rotatable with respect to the body, wherein the electric motor and the transmission unit are disposed in the first handle, and the battery pack is disposed at a bottom of the body, and when the power tool stands, an orthographic projection of a center of gravity of the power tool on the standing surface is in the standing surface, and wherein the clutch mechanism comprises a shaft sleeve and a moving member, the moving member is connected to the shaft sleeve, and the shaft sleeve is movably connected to the transmission unit and driven by the moving member to be connected to or disengaged from the output unit.

18. The power tool according to claim 17, wherein the transmission unit further comprises a first bevel gear and a second bevel gear which mesh with each other and the first bevel gear is coaxially connected to a shaft of the transmission.
